Reg. No. | | | | | | | | | | [MSCCH405]

FOURTH SEMESTER UNIVERSITY EXAMINATION, JUNE 2016

M.Sc. (CHEMISTRY)

APPLIED CHEMISTRY

Time : 03 Hours **Maximum Marks:80**

Instructions: 1. Questions 1 to 6 carry 15 marks each, answer any four.

2. Question 7 is compulsory, carries 20 marks.

1. **a.** What are carbon nanotubes (CNTs)? Explain their synthesis by solvothermal method.

b. Explain the synthesis and characterization techniques of grahene and mention its applications. **8+7**

2. **a.** Write a note on epoxy resins and ion-exchange resins.

b. Explain the applications of polycarbonates. **10+5**

3. **a.** Write a note on the activity of pigments with examples.

b. Give an account on the activity of congo red and malachite green dyes. **10+5**

4. **a.** Explain the contact process manufacturing of sulphuric acid. Mention its chemical properties and industrial applications.

b. What are nitro-phosphates? Explain their preparation method and importance. **10+5**

5. Write in detail about the extraction methods of gold, zirconium and titanium. **15**

6. Write a note on the treatment of waste water from petroleum refinery, pulp and paper industry. **15**

7. **Answer any *TEN* of the following.** **10×2=20**

a. How to synthesize nanoparticles by sol-gel method?

b. Define fullerenes and list out their types.

c. What are polymer initiators? Give any two examples.

d. Explain glass transition temperature of polymers.

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e. What is the chemistry behind color development in dyes?

f. Give any two examples of "brightening agents" and explain their importance.

g. Explain the properties and agricultural uses of ammonium phosphate.

h. Mention any two activities of urea.

i. List out the applications of tungsten.

j. Define zone refining.

k. Write a short note on demineralization of water.

l. What are the primary contaminants present in waste water generated from fertilizer industry?